

LONGYAN, China

WMO: 589270

Lat: 25.100N

Lon: 117.017E

Elev: 341

StdP: 97.29

Time Zone: 8.00 (E08)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	4.4	-6.1	2.3	10.6	-3.5	2.9	9.2	6.2	12.5	5.4	11.9	1.4	0	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	34.5	24.1	33.6	24.1	32.7	24.0	25.5	31.0	25.1	30.5	24.8	30.0	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.9	27.2	23.8	19.4	27.0	23.5	19.1	26.8	80.4	31.1	79.0	30.7	77.7	30.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.9	4.3	DB	-0.1	36.8	1.7	0.9	-1.3	37.5	-2.3	38.0	-3.3	38.5	-4.6	39.1	
			WB	-1.6	27.0	1.5	1.0	-2.7	27.7	-3.6	28.3	-4.5	28.9	-5.6	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	12.5	13.3	16.1	20.1	23.8	26.3	28.3	27.9	25.9	22.6	17.8	13.2
	DBStd	6.45	3.58	3.74	4.28	3.54	2.73	2.18	1.55	1.47	2.22	2.73	3.21	3.57
	HDD10.0	43	18	8	4	0	0	0	0	0	0	0	1	12
	HDD18.3	656	183	144	94	22	2	0	0	0	0	2	48	161
	CDD10.0	3952	93	102	194	304	428	491	566	556	478	392	234	113
	CDD18.3	1524	1	4	26	76	172	241	308	297	228	136	31	3
	CDH23.3	12859	2	16	99	388	1111	2104	3321	2987	1837	836	148	10
(13)	CDH26.7	4676	0	1	9	81	327	706	1446	1228	630	231	18	0
(14)	Wind	WSAvg	1.6	1.6	1.5	1.6	1.5	1.6	1.7	1.8	1.6	1.6	1.7	1.4
(15) Precipitation	PrecAvg	1713	55	93	179	200	292	301	130	181	135	59	45	34
	PrecMax	2498	223	351	571	491	560	556	343	395	425	181	126	118
	PrecMin	1221	0	0	11	60	56	91	41	27	29	0	0	0
	PrecStd	294	60	76	120	105	120	124	64	93	88	47	39	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	25.2	27.9	31.0	32.9	34.6	36.4	35.2	34.2	32.9	28.7	24.7
		MCWB	16.4	16.8	19.9	21.5	23.2	24.3	24.2	23.8	24.5	23.0	20.4	16.7
	2%	DB	21.7	23.4	26.2	28.9	31.2	33.0	34.9	34.2	32.6	30.9	26.8	22.7
		MCWB	15.1	16.6	19.5	21.2	22.9	24.1	24.2	24.0	24.0	22.1	18.8	15.4
	5%	DB	20.1	21.5	24.4	27.2	30.0	31.8	33.8	33.3	31.5	29.2	25.0	21.0
		MCWB	14.4	15.7	19.0	20.7	22.6	24.1	24.2	23.9	23.5	21.2	18.2	14.3
	10%	DB	18.3	19.5	22.4	25.6	28.5	30.5	32.8	32.3	30.3	27.7	23.2	19.3
		MCWB	13.5	15.3	18.3	20.4	22.3	23.9	24.3	24.0	23.0	20.1	16.9	13.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.9	21.2	23.2	24.7	25.7	25.9	26.0	25.4	23.9	21.5	17.6
		MCDB	20.9	22.9	25.9	28.5	29.1	30.4	32.1	31.7	31.3	30.9	26.7	22.7
	2%	WB	16.4	17.7	20.3	22.3	24.0	25.2	25.3	25.4	25.0	23.1	20.3	16.5
		MCDB	20.3	21.3	24.2	26.8	28.7	29.9	31.3	30.9	30.4	29.2	24.5	21.2
	5%	WB	15.3	16.9	19.6	21.8	23.6	24.9	25.0	25.0	24.5	22.4	19.4	15.4
		MCDB	19.0	19.9	23.2	25.9	27.9	29.4	30.9	30.4	29.5	27.3	23.1	19.5
	10%	WB	14.2	16.0	18.9	21.2	23.1	24.5	24.7	24.6	24.1	21.7	18.4	14.4
		MCDB	17.5	18.9	22.2	24.8	27.0	28.8	30.2	29.7	28.5	25.7	22.2	18.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	6.7	6.2	6.5	6.8	6.7	8.5	8.3	7.5	8.6	9.1	9.8
		MCDBR	11.3	10.9	9.6	9.2	9.5	8.9	10.7	10.2	9.6	10.7	11.2	12.0
		MCWBR	5.5	5.1	4.0	3.3	3.0	2.3	2.5	2.4	2.6	3.4	4.4	5.4
	5% WB	MCDBR	8.8	8.0	7.7	7.2	7.5	7.1	8.8	8.5	8.2	9.2	9.1	9.5
		MCWBR	4.6	4.3	3.7	3.1	2.8	2.4	2.5	2.6	2.5	3.3	4.0	4.5
(40) Clear-Sky Solar Irradiance	taub		0.568	0.638	0.753	0.744	0.622	0.576	0.550	0.616	0.630	0.617	0.544	0.527
	taud		1.760	1.658	1.484	1.546	1.823	1.944	2.014	1.824	1.771	1.744	1.875	1.870
	Ebn at Noon		670	658	610	634	716	743	763	712	688	665	688	686
	Edh at Noon		201	238	297	285	215	189	176	213	219	215	177	173
(44) All-Sky Solar Radiation	RadAvg		2.73	2.85	3.12	3.52	3.86	4.15	5.32	4.84	4.41	3.94	3.13	2.90
	RadStd		0.46	0.54	0.47	0.56	0.62	0.50	0.47	0.46	0.40	0.59	0.40	0.48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.32	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	+110	+89

Nomenclature: See separate page